Supplement S3

Simulation study: coverage of the F1-score intervals

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1 Introduction

Supplement to the paper "Confidence intervals for the F1-score in medical object detection: comparing bootstrap and Bayesian multilevel methods".

This script evaluates the simulation study: coverage, bias and width of the 95% intervals of recall, precision and F1-score. The simulation itself (data generation, model fits, bootstrap, extraction of the data files) is in the folder `simulation/` of the repository.

Code and data: https://github.com/ndevln/f1-score-ci-object-detection (data: `S3_Data_ML_Estimates.csv.xz`, `S3_Data_Pathologist_Estimates.csv.xz`, `S3_Data_Objects_per_Image.csv`).

ML design (one rater per image, as the ML models of the examples): 4 image scenarios x 2 image counts (40, 100) x 500 replicates with 4 (A-C) or 3 (D) performance variants = 15,000 datasets.

- Scenarios (objects P per image): D very low (P = 1 + Poisson(0.5), median 1, mean 1.5), A low (median 4, 43% of the images with P <= 3), B broad (median 20), C bimodal (median 22, 20% large images).
- Variants of A-C: 1 base (recall 0.8, 2 FP per image + 0.05 FP per object, image SDs 0.8 / 1.2, correlation 0.7); 2 FP independent of P; 3 higher performance (recall 0.9, 1 FP per image); 4 low image heterogeneity and lower performance (recall 0.65, 3.1 FP per image, image SDs 0.3 / 0.4).
- Variants of D (fewer FP, as in radiology; image SDs 0.8 / 1.2, correlation 0.7, no FP per object): 5 baseline (recall 0.8, 0.1 FP per median image, about 0.2 FP per image on average); 6 more FP (0.5 FP per median image, about 1.0 on average); 7 conservative operating point (recall 0.65, 0.05 FP per median image, about 0.1 on average).
- Methods: image bootstrap (pooled and grouped); Bayesian binomial + Poisson model; Bayesian multinomial model as a comparison model (both models defined in the Appendix); Wilson interval of the pooled recall and precision (objects treated as independent, no F1).

Pathologist design (every pathologist scores every image, as in Example 1): 3 image scenarios (A-C) x 2 variants x 100 replicates = 600 datasets with 40 images.

- Variants: 1 “data example” (23 pathologists, mainly differing in their cutoff), 2 “skilled panel” (9 pathologists, mainly differing in skill).
- Average pathologist (the population of pathologists): row bootstrap (image x pathologist rows), cluster bootstrap (images with all pathologists), two-way bootstrap (images and pathologists); Bayesian binomial + Poisson model with new pathologists and with the observed pathologists (as in Example 1).
- Single pathologist (one new pathologist, prediction interval): stratified bootstrap (images within each pathologist), cluster bootstrap; Bayesian binomial + Poisson model.

Estimands and truths: marginal pooled and grouped values of the superpopulation of images (and of the population of pathologists), computed from expected counts. The estimates of the average of the *observed* pathologists (Example 1: cluster bootstrap and the Bayesian model with the observed pathologists) are compared with the population of pathologists as well, to show that they do not generalize to other pathologists.

Evaluation: 95% quantile intervals (QI); point estimate = median of the bootstrap or posterior draws (Wilson: observed proportion and Wilson score interval). Coverage = share of the datasets whose interval contains the truth; for the single pathologist the mean share of the 20,000 per-pathologist truths inside the interval. Monte Carlo SE of the coverage = sd / $\sqrt{n}$ over the n datasets of a table cell (for a share of 95% about 0.2 pp with 12,000 datasets, 0.5 pp with 2,000, 1.0 pp with 500 and 2.2 pp with 100). The cells (scenario x image count) are ordered by the expected number of objects per dataset (mean P x number of images), e.g. “D 40 (60)”.

2 Setup

2.1 Load Packages

```
library(readr)
library(dplyr)
library(tidyr)
library(purrr)
library(ggplot2)
```

2.2 Functions

`summarise_coverage()` summarises grouped datasets: coverage, its Monte Carlo SE, bias and width. `format_coverage()` prints the coverage as “93.6 (0.2)” (Monte Carlo SE in brackets) and rounds the other columns.

```
summarise_coverage <- function(df) {
  df |>
    summarise(
      datasets = n(),
      coverage = 100 * mean(covered),
      mc_se    = 100 * sd(covered) / sqrt(n()),
      bias     = mean(bias),
      width    = mean(width),
      .groups = "drop"
    )
}

format_coverage <- function(df) {
  df |>
    mutate(
      coverage = sprintf("%.1f (%.1f)", coverage, mc_se),
      across(c(bias, width), \(x) round(x, 3))
    ) |>
    select(-mc_se, -datasets)
}
```

2.3 Objects per Dataset

Expected number of objects per dataset: mean P of the superpopulation of a scenario times the number of images. It orders the cells and the scenarios (D, A, B, C) from few to many objects.

```
read_csv("S3_Data_Objects_per_Image.csv", show_col_types = FALSE) |>
  cross_join(tibble(n_slide = c(40, 100))) |>
  mutate(objects = round(mean_P * n_slide), mean_P = round(mean_P, 2)) |>
  arrange(objects) |>
  mutate(cell = sprintf("%s %d (%s)", scenario, n_slide,
                        format(objects, big.mark = ",", trim = TRUE)),
         cell = factor(cell, levels = cell)) -> df_cells

df_cells
```

scenario	mean_P	n_slide	objects	cell
D	1.50	40	60	D 40 (60)
D	1.50	100	150	D 100 (150)
A	4.40	40	176	A 40 (176)
A	4.40	100	440	A 100 (440)
B	31.19	40	1248	B 40 (1,248)
C	63.33	40	2533	C 40 (2,533)
B	31.19	100	3119	B 100 (3,119)
C	63.33	100	6333	C 100 (6,333)

3 ML design

3.1 Load Data

Estimates of all methods (95% QI and Wilson intervals), joined with the truths.

```
scenario_levels <- df_cells |> distinct(scenario, mean_P) |> arrange(mean_P) |> pull(scenario)

df_ml <- read_csv("S3_Data_ML_Estimates.csv.xz",
                  show_col_types = FALSE) |>
  left_join(df_cells |> select(scenario, n_slide, objects, cell), by = c("scenario", "n_slide")) |>
  mutate(
    scenario  = factor(scenario, levels = scenario_levels),
    scenarios = if_else(scenario == "D", "D (variants 5-7)", "A-C (variants 1-4)"),
    method = factor(method_id,
                    levels = c("bootstrap", "binomial_poisson", "multinomial",
                               "wilson"),
                    labels = c("Bootstrap (images resampled)", "Bayesian binomial + Poisson",
                               "Bayesian multinomial", "Wilson binomial CI")),
    # Short names for the tables
    method_short = factor(method, labels = c("Bootstrap", "Bayes B+P", "Bayes multinomial",
                                             "Wilson")),
    covered = as.numeric(.lower <= truth & truth <= .upper),
    bias    = value - truth,
    width   = .upper - .lower
  )
```

3.2 Coverage, bias and width

Coverage, bias and width of the intervals per metric, scenario and image count, ordered by the expected number of objects per dataset (all variants; A-C: 2,000, D: 1,500 datasets per point; Monte Carlo SE of the coverage about 0.5 pp near 95%, up to 1.2 pp where the coverage drops). Long format for the facets: one row per measure (coverage, bias, width).

```
measure_levels <- c("Coverage (%)", "Bias", "Width")

df_ml |>
  group_by(name, averaging, method, cell) |>
  summarise_coverage() |>
  pivot_longer(c(coverage, bias, width), names_to = "measure") |>
  mutate(measure = factor(measure, levels = c("coverage", "bias", "width"),
                          labels = measure_levels)) -> df_ml_cells
```

```r
# Reference lines: 95% coverage, zero bias
tibble(measure = factor(measure_levels[1:2], levels = measure_levels),
       value = c(95, 0)) -> df_reference

# Axis limits: coverage up to 100%, bias at least -0.02 to 0.02 (tiny biases do not look large)
tibble(measure = factor(measure_levels[c(1, 2, 2)], levels = measure_levels),
       value = c(100, -0.02, 0.02)) -> df_limits
```

Figure template without data and facets; one figure per metric is the template plus the data of the metric (`plot + df` replaces the plot data) and the facets with the metric in the row labels (e.g. "F1-score CI coverage (%)"), kept in a named list (Supplement Figures S3.1-S3.3; Figure 4 of the paper).

```r
ggplot(mapping = aes(x = cell, y = value, color = method, group = method)) +
  geom_hline(data = df_reference, aes(yintercept = value), linetype = "dashed") +
  geom_blank(data = df_limits, aes(y = value), inherit.aes = FALSE) +
  geom_line() +
  geom_point(size = 1) +
  # Wilson in vermillion (6th Okabe-Ito colour)
  ggokabeito::scale_color_okabe_ito(name = NULL, order = c(1, 2, 3, 6),
                                    guide = guide_legend(nrow = 2)) +
  scale_x_discrete(guide = guide_axis(angle = 90)) +
  labs(x = "Scenario, number of images (expected number of objects per dataset)", y = NULL) +
  theme_bw() +
  theme(legend.position = "top", panel.grid.minor = element_blank(),
        strip.placement = "outside") -> ggplot_template_ml

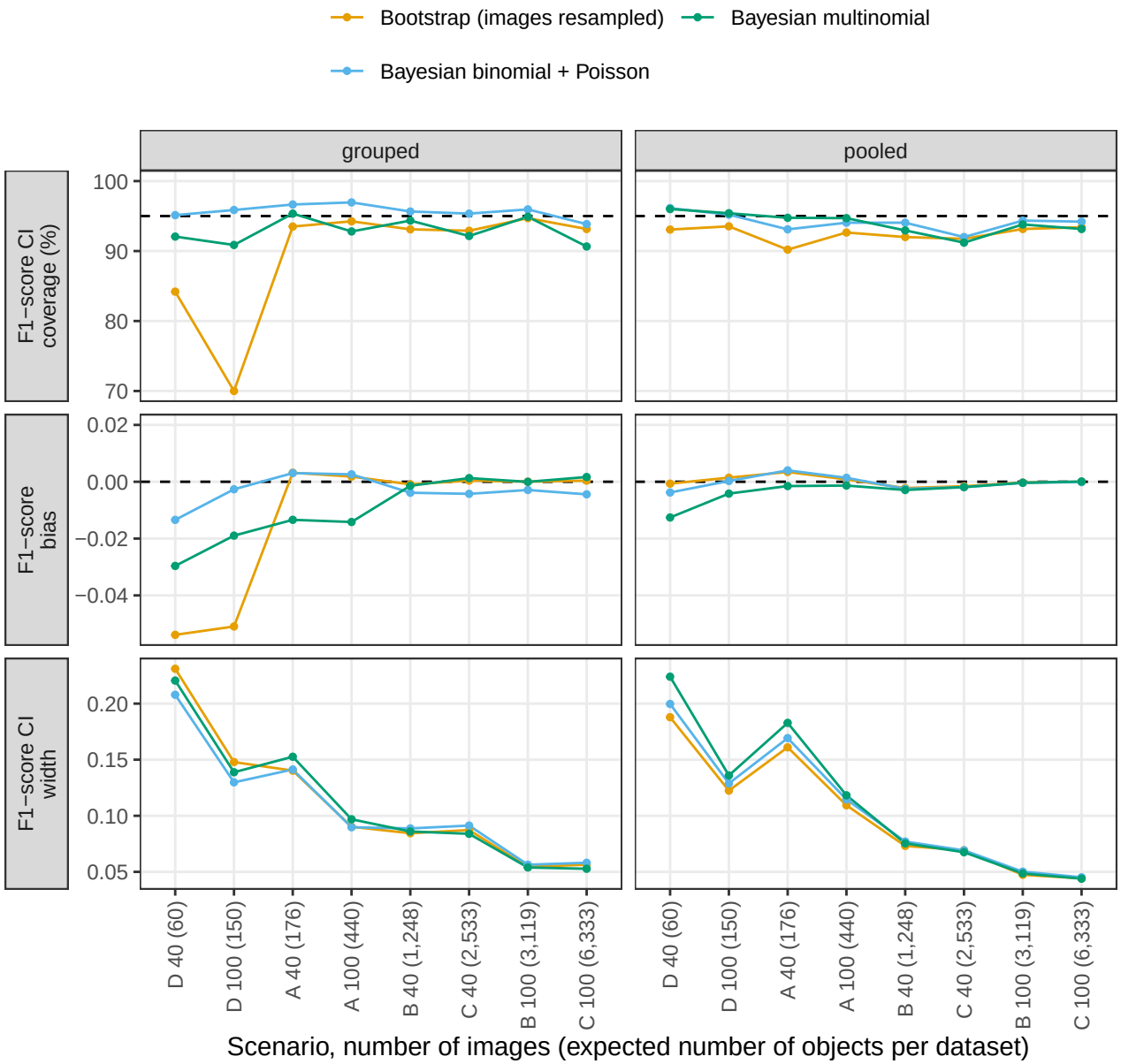
# Named list (F1, Precision, Recall)
df_ml_cells |>
  split(~name) |>
  imap(\(df, metric) {
    metric_label <- if_else(metric == "F1", "F1-score", metric)
    ggplot_template_ml + df +
      facet_grid(measure ~ averaging, scales = "free_y", switch = "y",
                 labeller = labeller(measure = \(x) case_match(
                   x,
                   "Coverage (%)" ~ paste(metric_label, "CI\ncoverage (%)"),
                   "Bias"         ~ paste0(metric_label, "\nbias"),   # bias of the point estimate
                   "Width"        ~ paste(metric_label, "CI\nwidth")
                 )))
  }) -> plots_ml_cells
```

3.2.1 Supplement Figure S3.1 (Figure 4)

F1-score with all methods. Figure 4 of the paper shows the same without the multinomial model (comparison model); it is saved as `Figure_4.pdf`.

```r
plots_ml_cells$F1
```

```
Warning: `case_match()` was deprecated in dplyr 1.2.0.
i Please use `recode_values()` instead.
```



Supplement Figure S3.1: Coverage, bias and mean width of the 95% intervals of the F1-score in the ML design per scenario and image count, ordered by the expected number of objects per dataset (in brackets); all variants (A-C: 2,000, D: 1,500 datasets per point). Dashed: 95% coverage and zero bias. Figure 4 of the paper shows the same without the multinomial model.

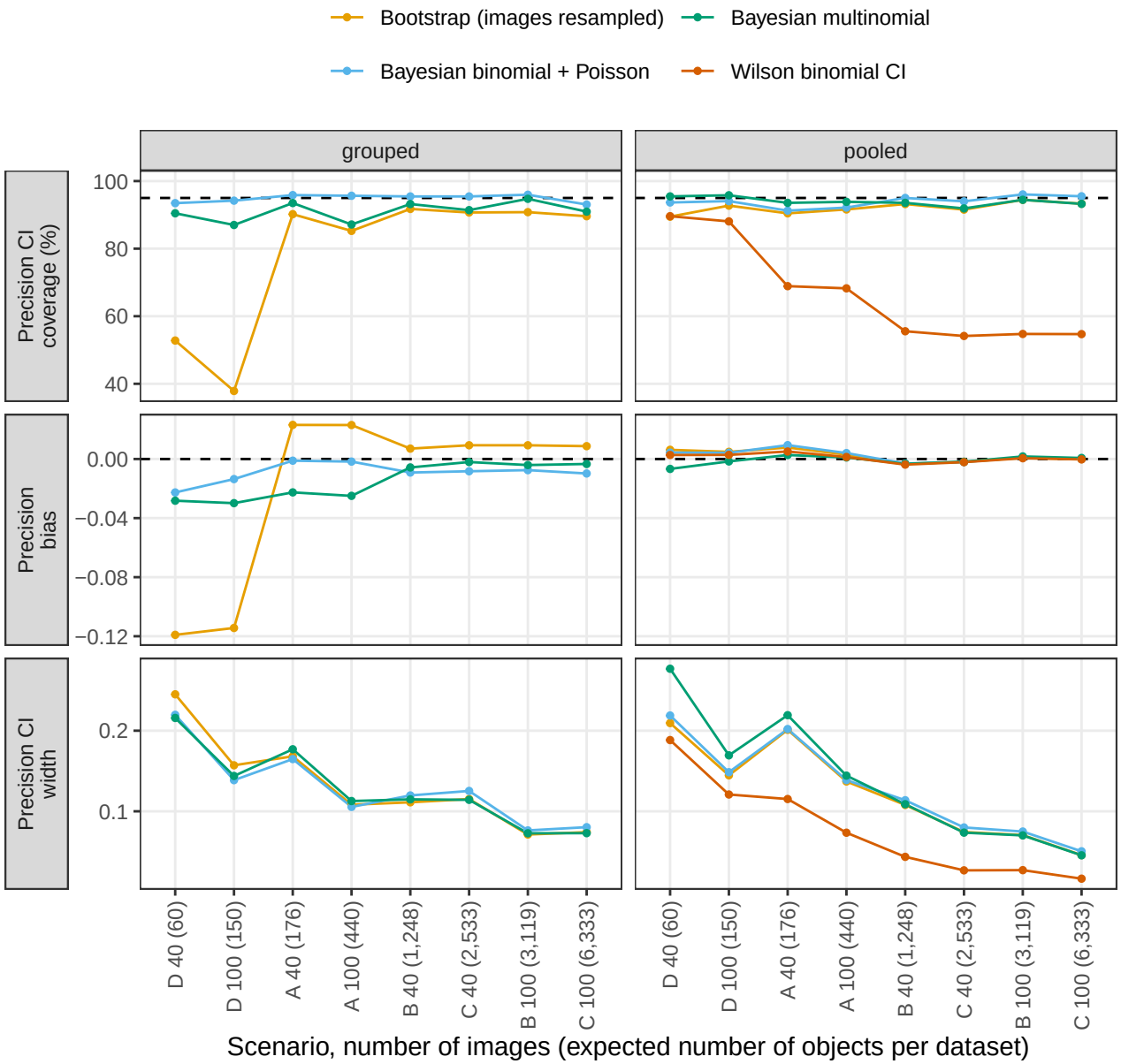
```
# Figure 4 of the paper: without the multinomial model
plots_ml_cells$F1 +
  filter(df_ml_cells, name == "F1", method != "Bayesian multinomial") +
  guides(color = guide_legend(nrow = 1)) -> fig_4

ggsave(fig_4, filename = "Figure_4.pdf", width = 7, height = 5.5)
```

3.2.2 Supplement Figures S3.2 and S3.3

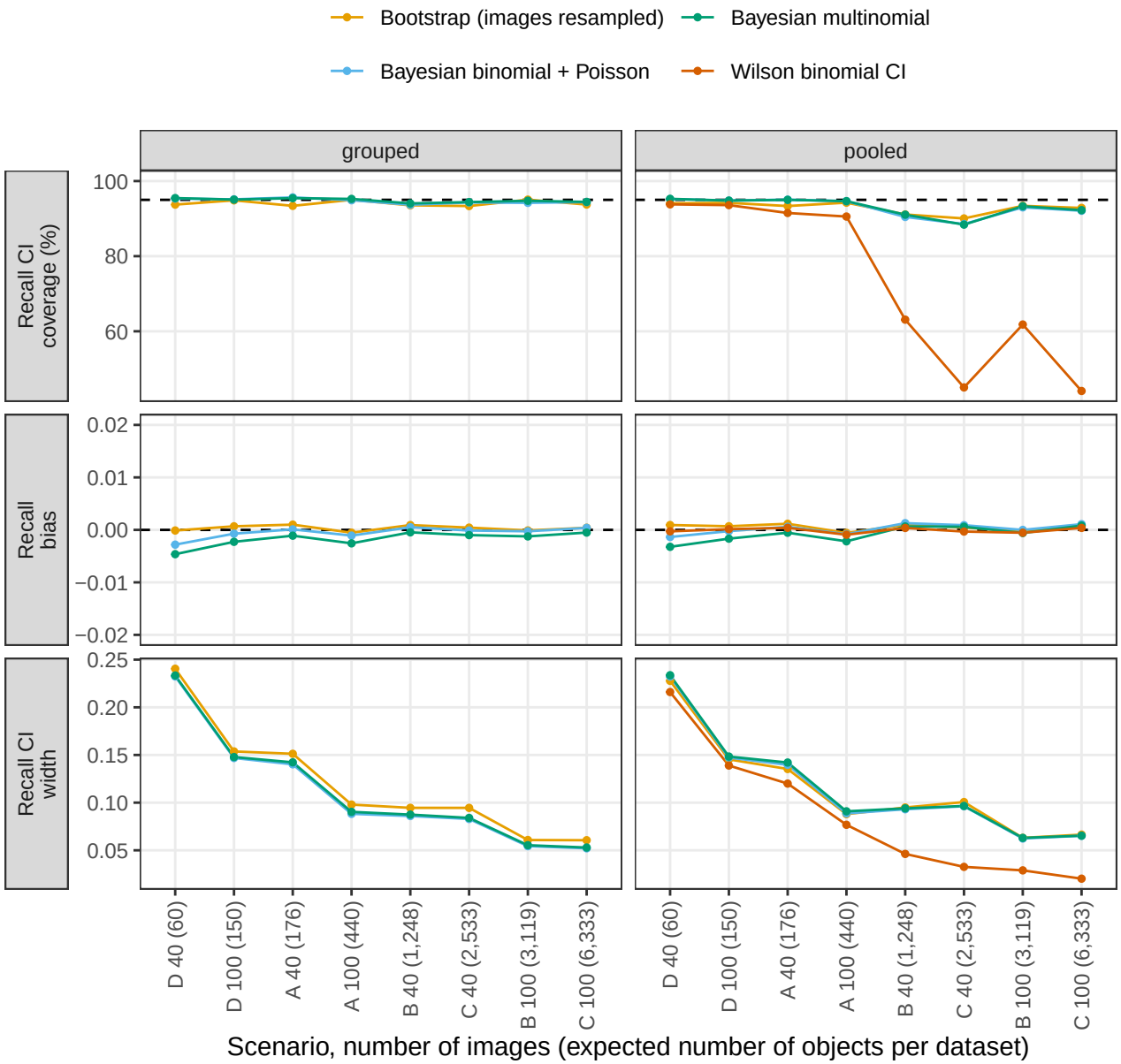
Precision and recall: the same, with the Wilson interval for the pooled estimand. The Wilson interval treats all objects as independent; its width shrinks with the number of objects, while the uncertainty depends on the number of images.

```
plots_ml_cells$Precision
```



Supplement Figure S3.2: Coverage, bias and mean width of the 95% intervals of the precision in the ML design, as in Supplement Figure S3.1; Wilson interval for the pooled estimand only.

```
plots_ml_cells$Recall
```



Supplement Figure S3.3: Coverage, bias and mean width of the 95% intervals of the recall in the ML design, as in Supplement Figure S3.1; Wilson interval for the pooled estimand only.

3.3 Overview

Coverage in % (Monte Carlo SE), mean bias and width: scenarios A-C (12,000 datasets) and D (3,000 datasets).

```
df_ml |>
  group_by(scenarios, name, averaging, method = method_short) |>
  summarise_coverage() |>
  format_coverage()
```

scenarios	name	averaging	method	coverage	bias	width
A-C (variants 1-4)	F1	grouped	Bootstrap	93.6 (0.2)	0.001	0.086
A-C (variants 1-4)	F1	grouped	Bayes B+P	95.7 (0.2)	-0.002	0.088
A-C (variants 1-4)	F1	grouped	Bayes multinomial	93.4 (0.2)	-0.004	0.088
A-C (variants 1-4)	F1	pooled	Bootstrap	92.2 (0.2)	0.000	0.084
A-C (variants 1-4)	F1	pooled	Bayes B+P	93.6 (0.2)	0.000	0.088
A-C (variants 1-4)	F1	pooled	Bayes multinomial	93.4 (0.2)	-0.001	0.089
A-C (variants 1-4)	Precision	grouped	Bootstrap	89.7 (0.3)	0.013	0.108

scenarios	name	averaging	method	coverage	bias	width
A-C (variants 1-4)	Precision	grouped	Bayes B+P	95.2 (0.2)	-0.006	0.112
A-C (variants 1-4)	Precision	grouped	Bayes multinomial	91.8 (0.3)	-0.010	0.111
A-C (variants 1-4)	Precision	pooled	Bootstrap	92.4 (0.2)	0.001	0.106
A-C (variants 1-4)	Precision	pooled	Bayes B+P	94.0 (0.2)	0.002	0.110
A-C (variants 1-4)	Precision	pooled	Bayes multinomial	93.4 (0.2)	0.000	0.110
A-C (variants 1-4)	Precision	pooled	Wilson	59.4 (0.4)	0.000	0.050
A-C (variants 1-4)	Recall	grouped	Bootstrap	94.0 (0.2)	0.000	0.093
A-C (variants 1-4)	Recall	grouped	Bayes B+P	94.5 (0.2)	0.000	0.084
A-C (variants 1-4)	Recall	grouped	Bayes multinomial	94.7 (0.2)	-0.001	0.085
A-C (variants 1-4)	Recall	pooled	Bootstrap	92.5 (0.2)	0.000	0.091
A-C (variants 1-4)	Recall	pooled	Bayes B+P	92.3 (0.2)	0.001	0.091
A-C (variants 1-4)	Recall	pooled	Bayes multinomial	92.4 (0.2)	0.000	0.092
A-C (variants 1-4)	Recall	pooled	Wilson	66.0 (0.4)	0.000	0.054
D (variants 5-7)	F1	grouped	Bootstrap	77.1 (0.8)	-0.052	0.189
D (variants 5-7)	F1	grouped	Bayes B+P	95.5 (0.4)	-0.008	0.169
D (variants 5-7)	F1	grouped	Bayes multinomial	91.5 (0.5)	-0.024	0.180
D (variants 5-7)	F1	pooled	Bootstrap	93.3 (0.5)	0.000	0.155
D (variants 5-7)	F1	pooled	Bayes B+P	95.7 (0.4)	-0.002	0.164
D (variants 5-7)	F1	pooled	Bayes multinomial	95.7 (0.4)	-0.008	0.180
D (variants 5-7)	Precision	grouped	Bootstrap	45.3 (0.9)	-0.117	0.201
D (variants 5-7)	Precision	grouped	Bayes B+P	93.8 (0.4)	-0.018	0.179
D (variants 5-7)	Precision	grouped	Bayes multinomial	88.7 (0.6)	-0.029	0.180
D (variants 5-7)	Precision	pooled	Bootstrap	91.1 (0.5)	0.006	0.177
D (variants 5-7)	Precision	pooled	Bayes B+P	93.9 (0.4)	0.004	0.184
D (variants 5-7)	Precision	pooled	Bayes multinomial	95.6 (0.4)	-0.004	0.223
D (variants 5-7)	Precision	pooled	Wilson	88.8 (0.6)	0.003	0.155
D (variants 5-7)	Recall	grouped	Bootstrap	94.3 (0.4)	0.000	0.197
D (variants 5-7)	Recall	grouped	Bayes B+P	95.2 (0.4)	-0.002	0.190
D (variants 5-7)	Recall	grouped	Bayes multinomial	95.3 (0.4)	-0.003	0.191
D (variants 5-7)	Recall	pooled	Bootstrap	94.1 (0.4)	0.001	0.187
D (variants 5-7)	Recall	pooled	Bayes B+P	95.0 (0.4)	-0.001	0.190
D (variants 5-7)	Recall	pooled	Bayes multinomial	95.0 (0.4)	-0.002	0.191
D (variants 5-7)	Recall	pooled	Wilson	93.7 (0.4)	0.000	0.177

Bootstrap: bootstrap with images resampled; Bayes B+P: Bayesian binomial + Poisson model; Bayes multinomial: Bayesian multinomial model; Wilson: Wilson binomial CI (pooled recall and precision).

3.4 F1-score per variant

Coverage in %: variants v1-v4 of scenarios A-C (3,000 datasets each), v5-v7 of scenario D (1,000 datasets each).

```
df_ml |>
  filter(name == "F1") |>
  group_by(averaging, method = method_short, variant) |>
  summarise_coverage() |>
  mutate(variant = paste0("v", variant),
         coverage = sprintf("%.1f", coverage)) |>
  select(averaging, method, variant, coverage) |>
  pivot_wider(names_from = variant, values_from = coverage)
```

averaging	method	v1	v2	v3	v4	v5	v6	v7
grouped	Bootstrap	94.2	93.7	92.4	94.0	79.6	93.1	58.6
grouped	Bayes B+P	96.5	95.9	95.3	95.2	96.1	96.4	94.0
grouped	Bayes multinomial	93.0	94.5	92.2	93.9	94.3	87.0	93.1
pooled	Bootstrap	90.8	92.3	93.2	92.4	93.1	93.1	93.7
pooled	Bayes B+P	92.2	93.1	94.5	94.7	96.1	95.4	95.5
pooled	Bayes multinomial	91.9	93.5	94.1	94.2	96.0	96.3	94.8

Bootstrap: bootstrap with images resampled; Bayes B+P: Bayesian binomial + Poisson model; Bayes multinomial: Bayesian multinomial model.

4 Pathologist design

4.1 Load Data

Estimates of all methods (95% QI), joined with the truths. For the single pathologist the share of the per-pathologist truths inside the interval replaces the comparison with one truth.

```
df_path <- read_csv("S3_Data_Pathologist_Estimates.csv.xz",
                    show_col_types = FALSE) |>
  mutate(
    method = factor(method_id,
                    levels = c("bootstrap_row", "bootstrap_stratified", "bootstrap_cluster",
                               "bootstrap_two_way", "binomial_poisson",
                               "binomial_poisson_observed"),
                    labels = c("Row bootstrap", "Stratified bootstrap", "Cluster bootstrap",
                               "Two-way bootstrap",
                               "Bayesian binomial + Poisson (new pathologists)",
                               "Bayesian binomial + Poisson (observed pathologists)")),
    # Short names for the tables
    method_short = factor(method, labels = c("Row bootstrap", "Stratified bootstrap",
                                             "Cluster bootstrap", "Two-way bootstrap",
                                             "Bayes B+P (new)", "Bayes B+P (obs.)")),
    single  = estimand == "single_pathologist",
    covered = if_else(single, share_inside, as.numeric(.lower <= truth & truth <= .upper)),
    bias    = value - truth,   # NA for the single pathologist (no single truth)
    width   = .upper - .lower
  )
```

4.2 Average pathologist

Coverage of the population value of the pathologists in % (Monte Carlo SE), mean bias and width, 300 datasets per variant.

```
df_path |>
  filter(!single) |>
  group_by(name, averaging, variant, method = method_short) |>
  summarise_coverage() |>
  format_coverage()
```

name	averaging	variant	method	coverage	bias	width
F1	grouped	1	Row bootstrap	33.7 (2.7)	0.001	0.030
F1	grouped	1	Cluster bootstrap	85.7 (2.0)	0.001	0.100
F1	grouped	1	Two-way bootstrap	93.0 (1.5)	0.001	0.126
F1	grouped	1	Bayes B+P (new)	95.0 (1.3)	-0.002	0.131
F1	grouped	1	Bayes B+P (obs.)	85.3 (2.0)	0.003	0.101
F1	grouped	2	Row bootstrap	35.0 (2.8)	-0.002	0.051
F1	grouped	2	Cluster bootstrap	57.3 (2.9)	-0.002	0.092
F1	grouped	2	Two-way bootstrap	90.7 (1.7)	-0.001	0.223
F1	grouped	2	Bayes B+P (new)	95.7 (1.2)	-0.012	0.243
F1	grouped	2	Bayes B+P (obs.)	59.0 (2.8)	-0.003	0.094
F1	pooled	1	Row bootstrap	31.3 (2.7)	0.004	0.044
F1	pooled	1	Cluster bootstrap	76.0 (2.5)	0.004	0.134
F1	pooled	1	Two-way bootstrap	90.3 (1.7)	0.007	0.179
F1	pooled	1	Bayes B+P (new)	94.3 (1.3)	-0.005	0.207
F1	pooled	1	Bayes B+P (obs.)	75.0 (2.5)	0.004	0.128
F1	pooled	2	Row bootstrap	32.7 (2.7)	0.003	0.067
F1	pooled	2	Cluster bootstrap	58.0 (2.9)	0.003	0.124
F1	pooled	2	Two-way bootstrap	88.0 (1.9)	0.007	0.264
F1	pooled	2	Bayes B+P (new)	94.3 (1.3)	-0.024	0.386
F1	pooled	2	Bayes B+P (obs.)	56.3 (2.9)	0.003	0.119
Precision	grouped	1	Row bootstrap	31.3 (2.7)	0.009	0.038
Precision	grouped	1	Cluster bootstrap	83.3 (2.2)	0.010	0.128
Precision	grouped	1	Two-way bootstrap	93.0 (1.5)	0.010	0.175
Precision	grouped	1	Bayes B+P (new)	95.7 (1.2)	0.001	0.184

name	averaging	variant	method	coverage	bias	width
Precision	grouped	1	Bayes B+P (obs.)	83.7 (2.1)	0.002	0.130
Precision	grouped	2	Row bootstrap	36.7 (2.8)	0.007	0.060
Precision	grouped	2	Cluster bootstrap	60.0 (2.8)	0.007	0.116
Precision	grouped	2	Two-way bootstrap	91.0 (1.7)	0.008	0.254
Precision	grouped	2	Bayes B+P (new)	94.7 (1.3)	-0.012	0.272
Precision	grouped	2	Bayes B+P (obs.)	65.0 (2.8)	-0.006	0.120
Precision	pooled	1	Row bootstrap	27.0 (2.6)	0.006	0.058
Precision	pooled	1	Cluster bootstrap	71.0 (2.6)	0.006	0.172
Precision	pooled	1	Two-way bootstrap	88.3 (1.9)	0.011	0.249
Precision	pooled	1	Bayes B+P (new)	95.0 (1.3)	-0.002	0.279
Precision	pooled	1	Bayes B+P (obs.)	70.3 (2.6)	0.006	0.165
Precision	pooled	2	Row bootstrap	31.3 (2.7)	0.008	0.080
Precision	pooled	2	Cluster bootstrap	53.3 (2.9)	0.007	0.155
Precision	pooled	2	Two-way bootstrap	83.3 (2.2)	0.014	0.312
Precision	pooled	2	Bayes B+P (new)	95.7 (1.2)	-0.025	0.443
Precision	pooled	2	Bayes B+P (obs.)	53.3 (2.9)	0.008	0.149
Recall	grouped	1	Row bootstrap	33.7 (2.7)	0.003	0.033
Recall	grouped	1	Cluster bootstrap	78.0 (2.4)	0.003	0.085
Recall	grouped	1	Two-way bootstrap	95.0 (1.3)	0.003	0.133
Recall	grouped	1	Bayes B+P (new)	96.0 (1.1)	0.002	0.137
Recall	grouped	1	Bayes B+P (obs.)	79.3 (2.3)	0.003	0.085
Recall	grouped	2	Row bootstrap	42.7 (2.9)	0.003	0.049
Recall	grouped	2	Cluster bootstrap	66.0 (2.7)	0.003	0.083
Recall	grouped	2	Two-way bootstrap	94.7 (1.3)	0.004	0.170
Recall	grouped	2	Bayes B+P (new)	98.0 (0.8)	-0.001	0.192
Recall	grouped	2	Bayes B+P (obs.)	65.7 (2.7)	0.003	0.083
Recall	pooled	1	Row bootstrap	28.0 (2.6)	0.001	0.037
Recall	pooled	1	Cluster bootstrap	79.3 (2.3)	0.001	0.112
Recall	pooled	1	Two-way bootstrap	94.3 (1.3)	0.002	0.151
Recall	pooled	1	Bayes B+P (new)	94.0 (1.4)	0.000	0.153
Recall	pooled	1	Bayes B+P (obs.)	78.0 (2.4)	0.002	0.107
Recall	pooled	2	Row bootstrap	45.7 (2.9)	0.002	0.055
Recall	pooled	2	Cluster bootstrap	70.3 (2.6)	0.002	0.105
Recall	pooled	2	Two-way bootstrap	93.7 (1.4)	0.003	0.180
Recall	pooled	2	Bayes B+P (new)	98.3 (0.7)	-0.002	0.201
Recall	pooled	2	Bayes B+P (obs.)	68.7 (2.7)	0.003	0.101

Bayes B+P (new): Bayesian binomial + Poisson model with new pathologists; Bayes B+P (obs.): with the observed pathologists (as in Example 1).

4.3 Single pathologist

Mean share (%) of the 20,000 per-pathologist truths inside the interval (Monte Carlo SE) and the mean width, 300 datasets per variant.

```
df_path |>
  filter(single) |>
  group_by(name, averaging, variant, method = method_short) |>
  summarise_coverage() |>
  format_coverage() |>
  select(-bias)
```

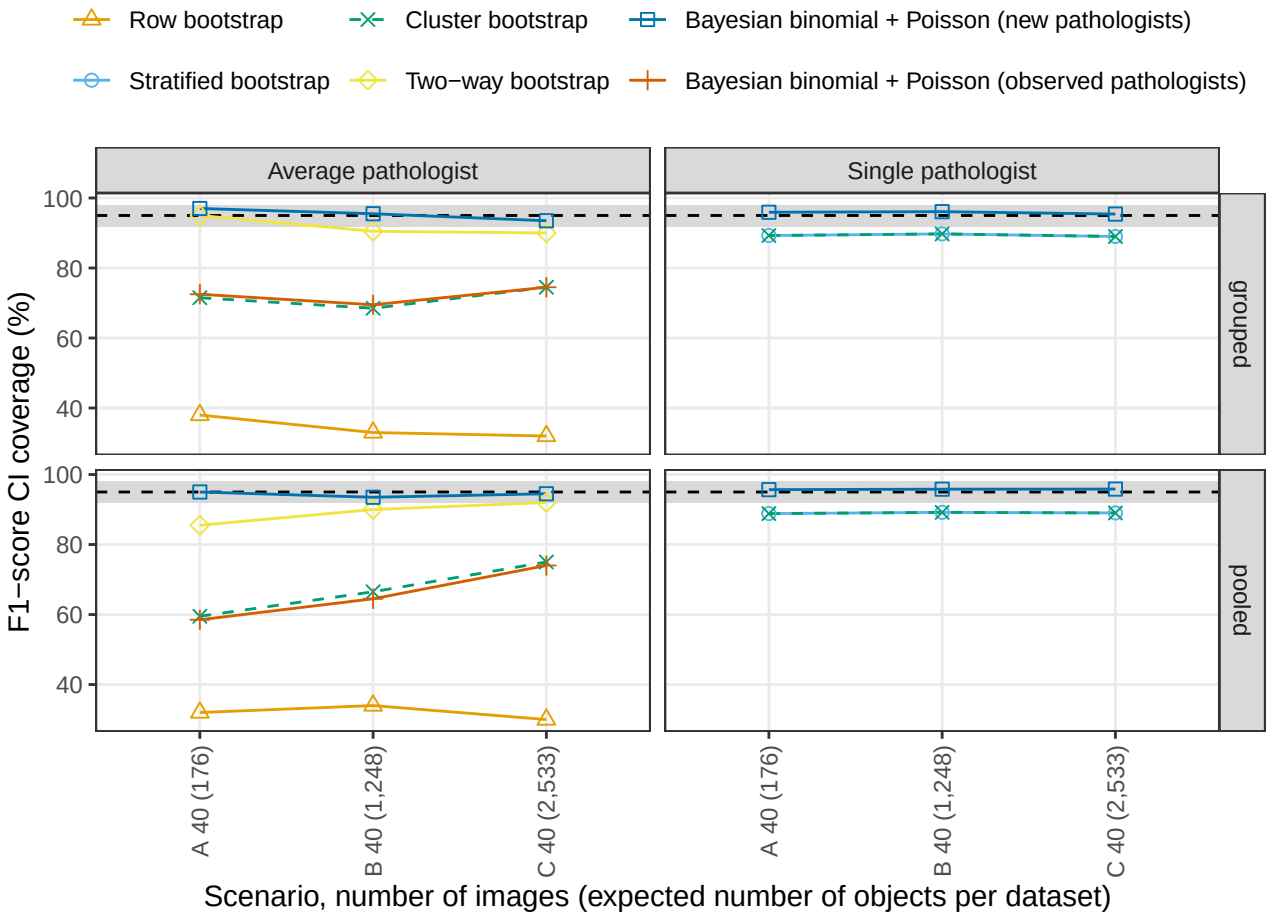
name	averaging	variant	method	coverage	width
F1	grouped	1	Stratified bootstrap	93.4 (0.3)	0.371
F1	grouped	1	Cluster bootstrap	93.4 (0.3)	0.371
F1	grouped	1	Bayes B+P (new)	95.8 (0.2)	0.401
F1	grouped	2	Stratified bootstrap	85.3 (0.6)	0.533
F1	grouped	2	Cluster bootstrap	85.3 (0.6)	0.533
F1	grouped	2	Bayes B+P (new)	95.8 (0.3)	0.692
F1	pooled	1	Stratified bootstrap	92.5 (0.4)	0.418
F1	pooled	1	Cluster bootstrap	92.5 (0.4)	0.418
F1	pooled	1	Bayes B+P (new)	95.4 (0.2)	0.451

name	averaging	variant	method	coverage	width
F1	pooled	2	Stratified bootstrap	85.5 (0.5)	0.537
F1	pooled	2	Cluster bootstrap	85.5 (0.5)	0.537
F1	pooled	2	Bayes B+P (new)	96.1 (0.3)	0.704
Precision	grouped	1	Stratified bootstrap	92.6 (0.3)	0.555
Precision	grouped	1	Cluster bootstrap	92.6 (0.3)	0.555
Precision	grouped	1	Bayes B+P (new)	95.6 (0.2)	0.600
Precision	grouped	2	Stratified bootstrap	86.5 (0.5)	0.592
Precision	grouped	2	Cluster bootstrap	86.5 (0.5)	0.592
Precision	grouped	2	Bayes B+P (new)	96.0 (0.3)	0.747
Precision	pooled	1	Stratified bootstrap	92.3 (0.3)	0.614
Precision	pooled	1	Cluster bootstrap	92.3 (0.3)	0.614
Precision	pooled	1	Bayes B+P (new)	96.0 (0.2)	0.655
Precision	pooled	2	Stratified bootstrap	85.0 (0.6)	0.607
Precision	pooled	2	Cluster bootstrap	85.1 (0.6)	0.607
Precision	pooled	2	Bayes B+P (new)	96.4 (0.3)	0.775
Recall	grouped	1	Stratified bootstrap	93.7 (0.3)	0.481
Recall	grouped	1	Cluster bootstrap	93.7 (0.3)	0.481
Recall	grouped	1	Bayes B+P (new)	95.6 (0.2)	0.503
Recall	grouped	2	Stratified bootstrap	88.7 (0.5)	0.406
Recall	grouped	2	Cluster bootstrap	88.7 (0.5)	0.406
Recall	grouped	2	Bayes B+P (new)	96.9 (0.3)	0.542
Recall	pooled	1	Stratified bootstrap	93.2 (0.3)	0.475
Recall	pooled	1	Cluster bootstrap	93.2 (0.3)	0.475
Recall	pooled	1	Bayes B+P (new)	95.6 (0.2)	0.507
Recall	pooled	2	Stratified bootstrap	88.1 (0.5)	0.402
Recall	pooled	2	Cluster bootstrap	88.1 (0.5)	0.402
Recall	pooled	2	Bayes B+P (new)	96.8 (0.3)	0.545

Bayes B+P (new): Bayesian binomial + Poisson model with one new pathologist.

Coverage of the F1-score intervals over both variants (200 datasets per point).

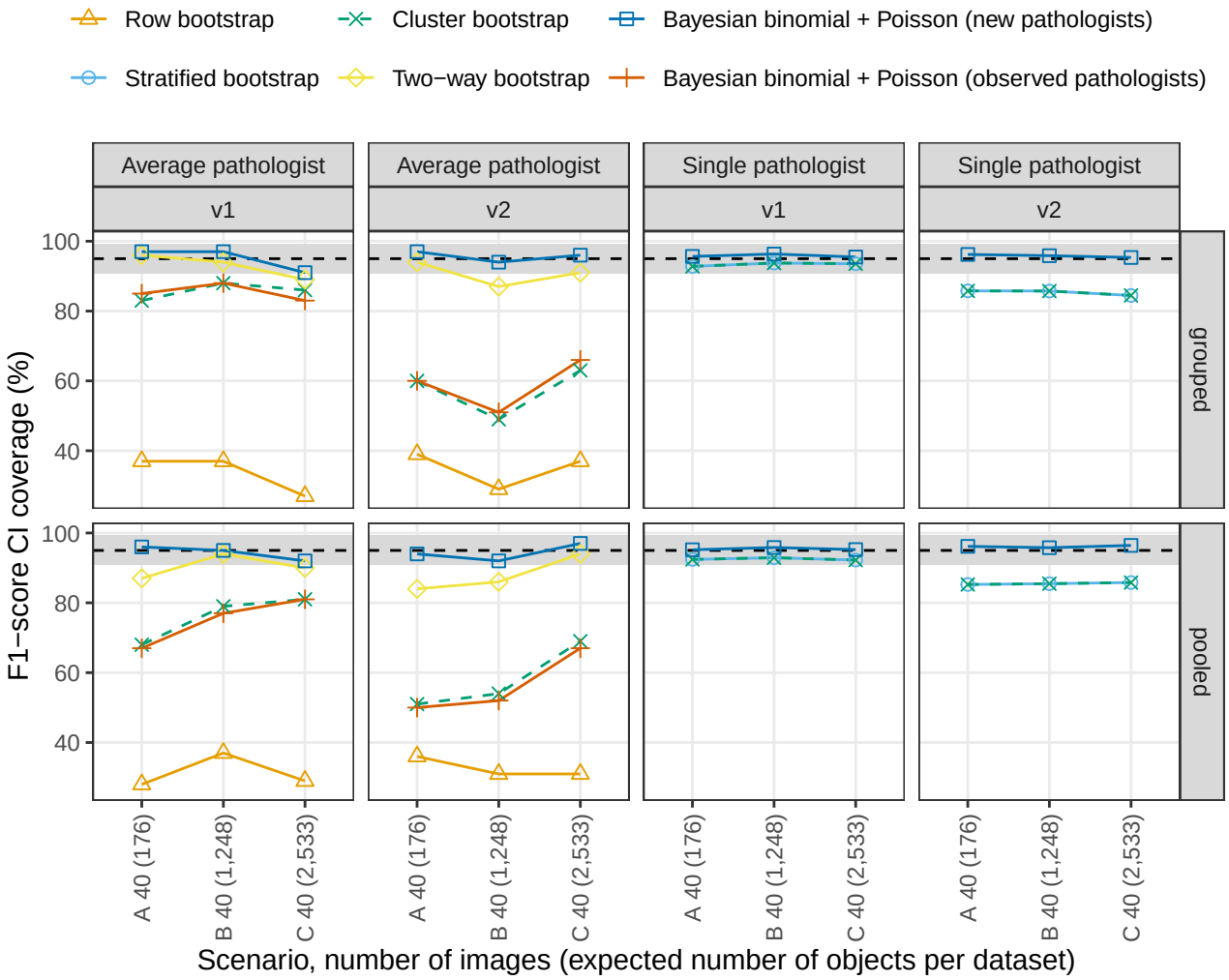
```
df_path |>
  filter(name == "F1") |>
  left_join(df_cells |> select(scenario, n_slide, cell), by = c("scenario", "n_slide")) |>
  group_by(estimand, averaging, method, cell) |>
  summarise_coverage() |>
  mutate(estimand = recode(estimand, average_pathologist = "Average pathologist",
                           single_pathologist = "Single pathologist")) |>
  ggplot(aes(x = cell, y = coverage, color = method, linetype = method, shape = method,
             group = method)) +
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = 95 - 1.96 * sqrt(95 * 5 / 200),
           ymax = 95 + 1.96 * sqrt(95 * 5 / 200), fill = "grey85") +
  geom_hline(yintercept = 95, linetype = "dashed") +
  geom_line() +
  geom_point(size = 2) +
  facet_grid(averaging ~ estimand) +
  ggokabeito::scale_color_okabe_ito(name = NULL, guide = guide_legend(nrow = 2)) +
  # Cluster bootstrap dashed: it lies exactly on the stratified bootstrap (single pathologist)
  scale_linetype_manual(name = NULL, values = c("solid", "solid", "dashed", "solid", "solid", "solid"),
                        guide = guide_legend(nrow = 2)) +
  scale_shape_manual(name = NULL, values = c(2, 1, 4, 5, 0, 3), guide = guide_legend(nrow = 2)) +
  scale_x_discrete(guide = guide_axis(angle = 90)) +
  labs(x = "Scenario, number of images (expected number of objects per dataset)", y = "F1-score CI coverage (%)") +
  theme_bw() +
  theme(legend.position = "top", panel.grid.minor = element_blank())
```



Supplement Figure S3.4: Coverage of the 95% intervals of the F1-score in the pathologist design per scenario (40 images), ordered by the expected number of objects per dataset (in brackets); both variants, 200 datasets per point. Single pathologist: mean share of the per-pathologist truths inside the interval. Dashed: 95%; grey band: 95% +- 1.96 Monte Carlo SE.

The same per variant (100 datasets per point).

```
df_path |>
  filter(name == "F1") |>
  left_join(df_cells |> select(scenario, n_slide, cell), by = c("scenario", "n_slide")) |>
  group_by(estimand, averaging, method, cell, variant) |>
  summarise_coverage() |>
  mutate(estimand = recode(estimand, average_pathologist = "Average pathologist",
                           single_pathologist = "Single pathologist"),
         variant = paste0("v", variant)) |>
  ggplot(aes(x = cell, y = coverage, color = method, linetype = method, shape = method,
             group = method)) +
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = 95 - 1.96 * sqrt(95 * 5 / 100),
           ymax = 95 + 1.96 * sqrt(95 * 5 / 100), fill = "grey85") +
  geom_hline(yintercept = 95, linetype = "dashed") +
  geom_line() +
  geom_point(size = 2) +
  facet_grid(averaging ~ estimand + variant) +
  ggokabeito::scale_color_okabe_ito(name = NULL, guide = guide_legend(nrow = 2)) +
  # Cluster bootstrap dashed: it lies exactly on the stratified bootstrap (single pathologist)
  scale_linetype_manual(name = NULL, values = c("solid", "solid", "dashed", "solid", "solid", "solid"),
                        guide = guide_legend(nrow = 2)) +
  scale_shape_manual(name = NULL, values = c(2, 1, 4, 5, 0, 3), guide = guide_legend(nrow = 2)) +
  scale_x_discrete(guide = guide_axis(angle = 90)) +
  labs(x = "Scenario, number of images (expected number of objects per dataset)", y = "F1-score CI coverage (%)") +
  theme_bw() +
  theme(legend.position = "top", panel.grid.minor = element_blank())
```



Supplement Figure S3.5: Coverage of the 95% intervals of the F1-score in the pathologist design as in Supplement Figure S3.4, per variant (v1: 23 pathologists, v2: 9 pathologists); 100 datasets per point. Dashed: 95%; grey band: 95% +- 1.96 Monte Carlo SE.

Appendix: Bayesian models

Binomial + Poisson model

The main model is the model of the paper (Eq. 6-12). For image i and subject j:

$$TP_{ij} \sim \text{Binomial}(n = P_i,\ p = recall_{ij}), \qquad FP_{ij} \sim \text{Poisson}(FPR_{ij})$$

$$\text{logit}(recall_{ij}) = \alpha_{recall} + u_{i,recall} + v_{j,recall}, \qquad \log(FPR_{ij}) = \alpha_{FPR} + u_{i,FPR} + v_{j,FPR}$$

with image random effects $u_{i,recall} \sim N(0, \sigma^2_{image,recall})$ and $u_{i,FPR} \sim N(0, \sigma^2_{image,FPR})$, and correlated subject random effects $(v_{j,recall}, v_{j,FPR}) \sim MVN(0, \Sigma_{person})$.

- ML design: one subject (the ML model) per dataset, so the model has no subject effects v_j.
- Pathologist design: the full model with the pathologists as subjects.
- Expected counts of image i: $TP_i = recall_{ij} \cdot P_i$, $FP_i = FPR_{ij}$.

Multinomial model

Comparison model of the ML design. The counts of image i are one multinomial outcome with TP as reference category:

$$(TP_i, FN_i, FP_i) \sim \text{Multinomial}\left(N_i = TP_i + FN_i + FP_i,\ (\pi_{i,TP}, \pi_{i,FN}, \pi_{i,FP})\right)$$

$$\log \frac{\pi_{i,FN}}{\pi_{i,TP}} = \alpha_{FN} + u_{i,FN}, \qquad \log \frac{\pi_{i,FP}}{\pi_{i,TP}} = \alpha_{FP} + \beta \widetilde{\log P_i} + u_{i,FP}$$

with $\widetilde{\log P_i}$ the log number of objects centred on the mean of the dataset and image random effects $u_{i,FN} \sim N(0, \sigma^2_{image,FN})$ and $u_{i,FP} \sim N(0, \sigma^2_{image,FP})$.

- Recall and precision of image i: $recall_i = \pi_{i,TP}/(\pi_{i,TP} + \pi_{i,FN})$, $precision_i = \pi_{i,TP}/(\pi_{i,TP} + \pi_{i,FP})$.
- Expected counts of image i: $TP_i = recall_i \cdot P_i$, $FP_i = TP_i \cdot \pi_{i,FP}/\pi_{i,TP}$.

Priors, sampling and estimates

- Priors: brms defaults, except the intercept of the FP part of the binomial + Poisson model: student_t(3, 1, 2.5) on α_{FPR}. The prior was fixed for the simulations, since the brms default is centred on the log median FP count of the data. Because the models were compiled once and refitted on every dataset, a data-dependent prior would be the one of the compile dataset.

Parameter	Binomial + Poisson	Multinomial
α_{recall}	student_t(3, 0, 2.5)	-
α_{FPR}	student_t(3, 1, 2.5) (fixed)	-
α_{FN}, α_{FP}	-	student_t(3, 0, 2.5)
β	-	flat
$\sigma_{image,\cdot}$	half-student_t(3, 0, 2.5)	half-student_t(3, 0, 2.5)
σ_{recall}, σ_{FPR} (pathologist design)	half-student_t(3, 0, 2.5)	-
ρ (pathologist design)	LKJ(1)	-

- Sampling: brms with cmdstanr, 4 chains with 4,000 iterations each, 1,000 warm-up, thinning 3 (4,000 posterior draws).
- Marginal pooled and grouped estimates as in the paper: per posterior draw the expected counts of the observed images, averaged with Bayesian bootstrap weights over the images (pooled: weighted sums of the counts, grouped: weighted mean of the per-image metrics); point estimate = median, 95% quantile interval.
- Pathologist design:
  - average observed pathologist (as in Example 1): their fitted subject random effects
  - average pathologist: the population of pathologists, 200 new pathologists from the subject random effects per draw
  - single pathologist: one new pathologist per draw
  - the Bayesian bootstrap weights are shared by all pathologists of an image